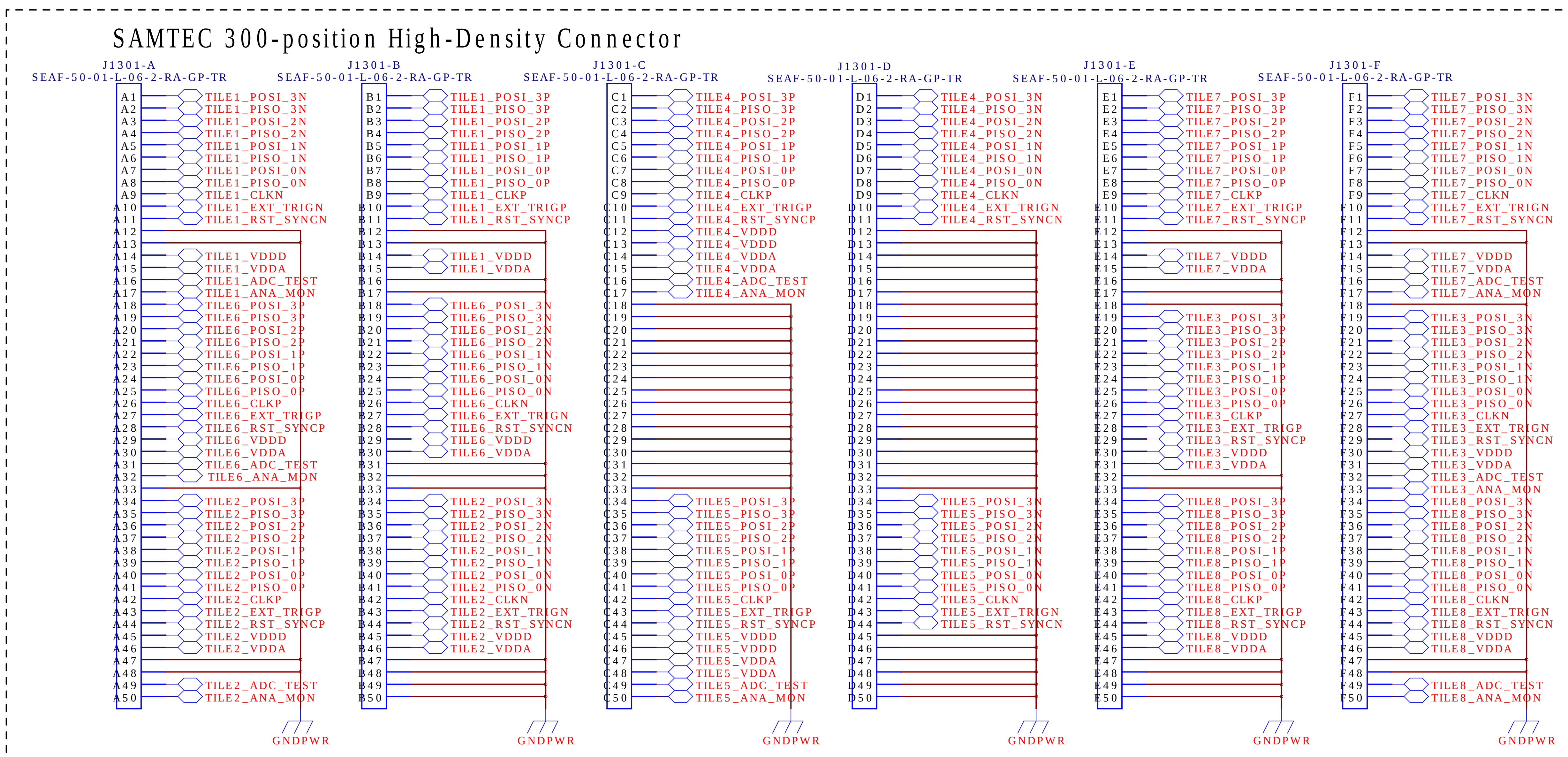
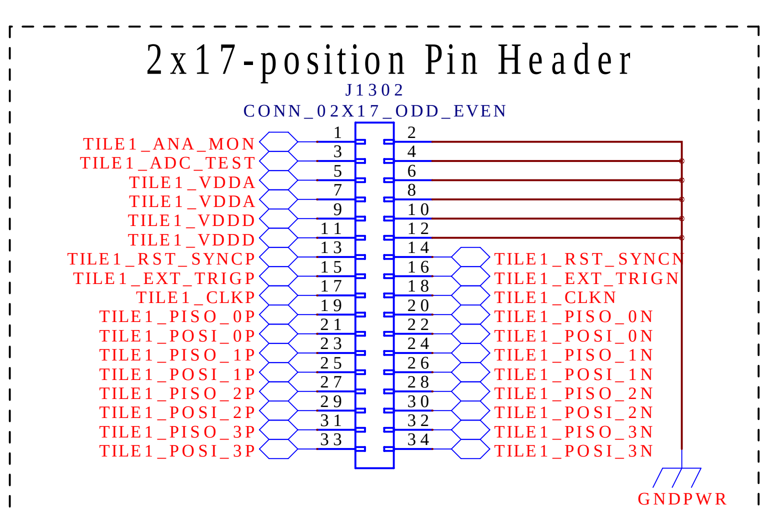




CONNECTOR

COMPANY:	University of California Davis
TITLE:	LarPix PACMAN Version 4

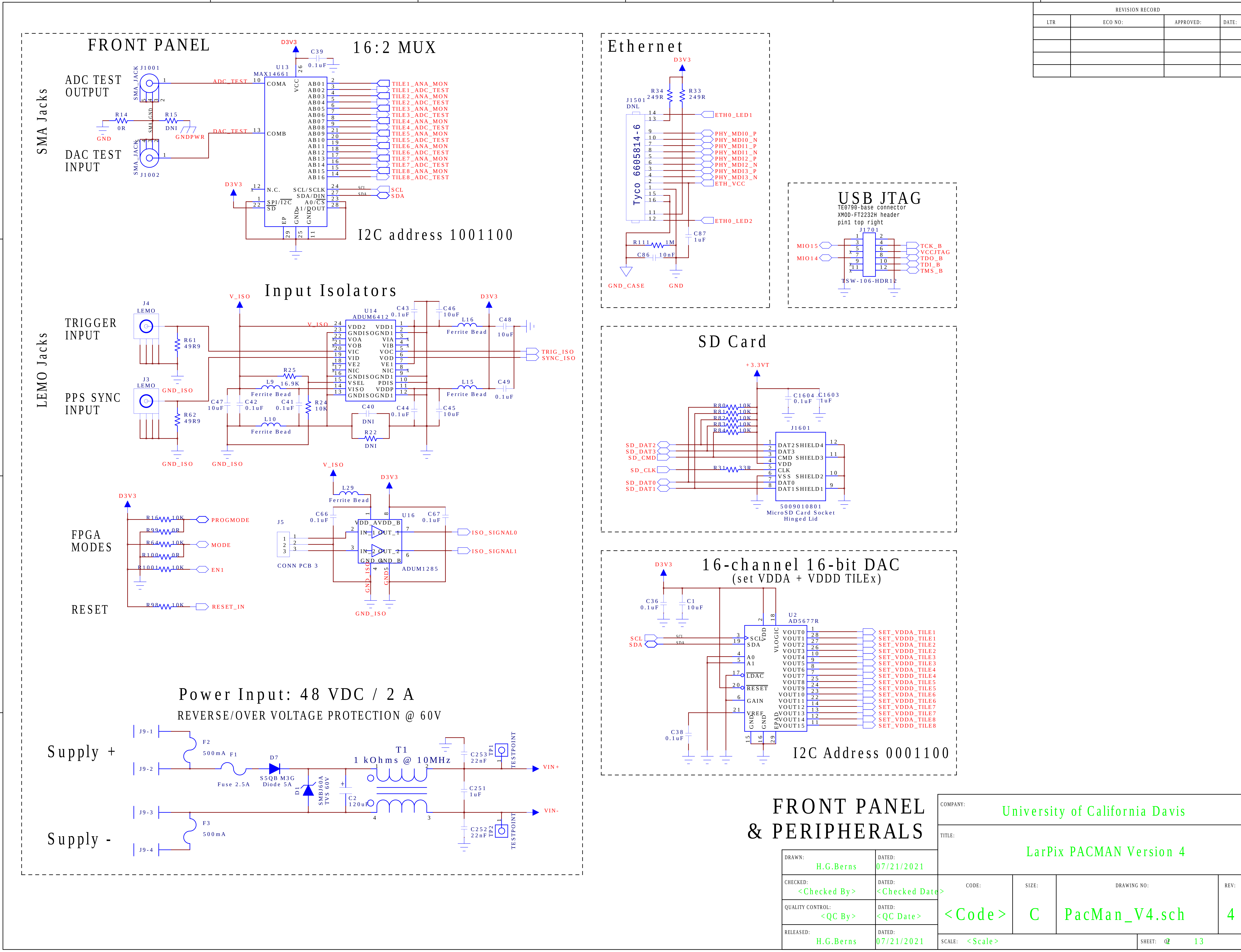
DRAWN: H.G.Berns	DATED: 07/21/2021		
CHECKED: <Checked By>	DATED: <Checked Date>		
QUALITY CONTROL: <QC By>	DATED: <QC Date>		
RELEASED: H.G.Berns	DATED: 07/21/2021		

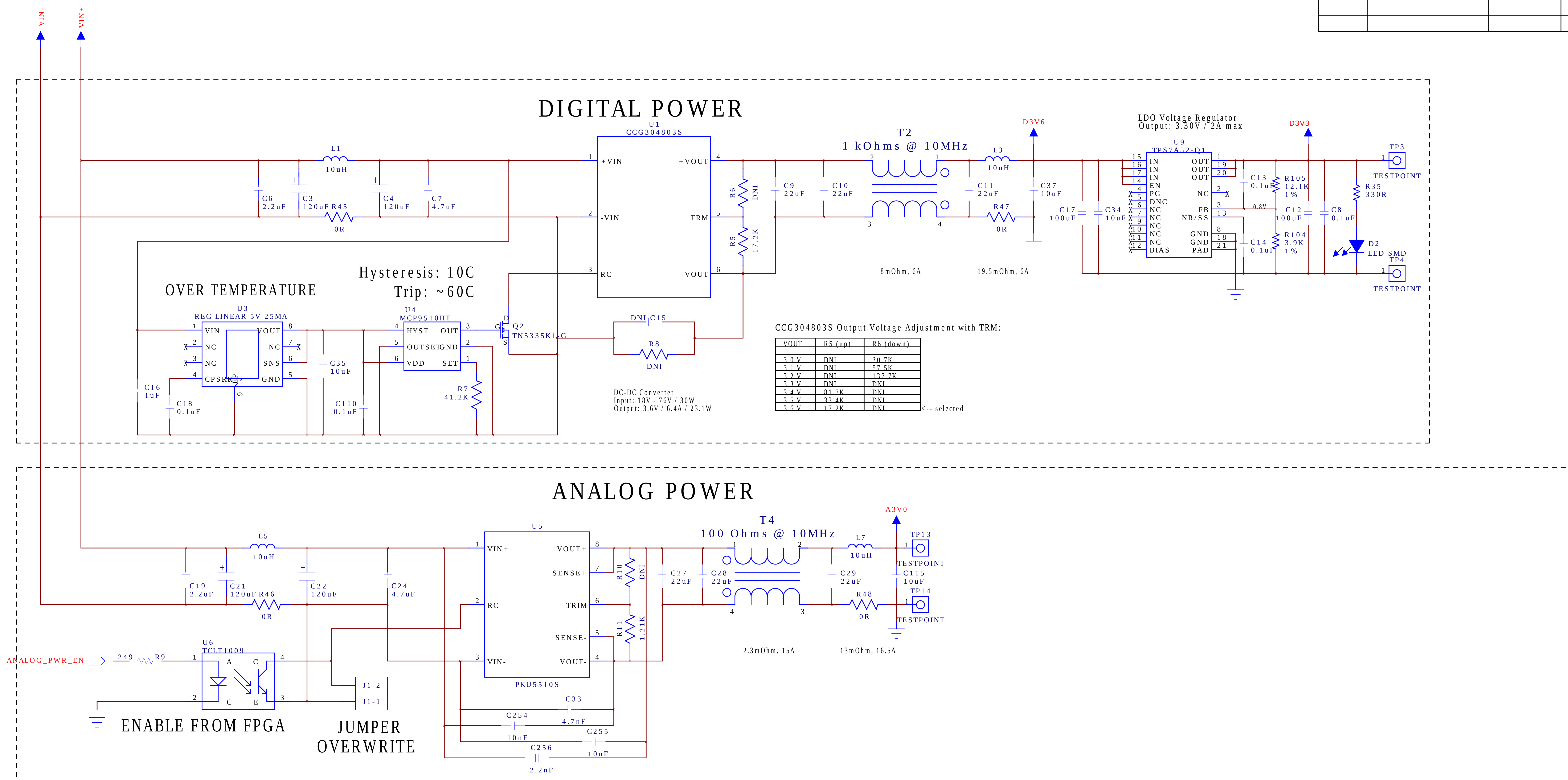
CODE:	SIZE:	DRAWING NO:	REV:
<Code>	C	PacMan_V4.sch	4

SCALE: <Scale> SHEET: 0 OF 13

REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:


FRONT PANEL
SMA Jacks
ADC TEST OUTPUT
DAC TEST INPUT
J1001
J1002
R14 0R
R15 DNI
GND
GNDPWR
ADC_TEST
DAC_TEST
16:2 MUX
U13 MAX14661
D3V3
C39 0.1uF
COMA
COMB
TILE1_ANA_MON
TILE1_ADC_TEST
TILE2_ANA_MON
TILE2_ADC_TEST
TILE3_ANA_MON
TILE3_ADC_TEST
TILE4_ANA_MON
TILE4_ADC_TEST
TILE5_ANA_MON
TILE5_ADC_TEST
TILE6_ANA_MON
TILE6_ADC_TEST
TILE7_ANA_MON
TILE7_ADC_TEST
TILE8_ANA_MON
TILE8_ADC_TEST
SCL
SDA
I2C address 1001100
Ethernet
J1501 DNL
Tyco 6605814-6
R34 249R
R33 249R
ETH0_LED1
PHY_MDI0_P
PHY_MDI0_N
PHY_MDI1_P
PHY_MDI1_N
PHY_MDI2_P
PHY_MDI2_N
PHY_MDI3_P
PHY_MDI3_N
ETH_VCC
ETH0_LED2
R111 1M
C86 10nF
C87 1uF
GND_CASE
GND
USB JTAG
TE0790-base connector
XMOD-FT2232H header
pin1 top right
J1701
MIO15
MIO14
TCK_B
VCCJTAG
TDO_B
TDI_B
TMS_B
TSW-106-HDR12
Input Isolators
LEMO Jacks
TRIGGER INPUT
PPS SYNC INPUT
J4 LEMO
J3 LEMO
R61 49R9
R62 49R9
V_ISO
GND_ISO
U14 ADUM6412
C43 0.1uF
C46 10uF
L16 Ferrite Bead
C48 10uF
TRIG_ISO
SYNC_ISO
R25 16.9K
L9 Ferrite Bead
C47 10uF
C42 0.1uF
C41 0.1uF
R24 10K
L10 Ferrite Bead
C40 DNI
R22 DNI
C44 0.1uF
C45 10uF
L15 Ferrite Bead
C49 0.1uF
SD Card
+3.3VT
C1604 0.1uF
C1603 1uF
R80 10K
R81 10K
R82 10K
R83 10K
R84 10K
R31 33R
J1601
SD_DAT2
SD_DAT3
SD_CMD
SD_CLK
SD_DAT0
SD_DAT1
5009010801
MicroSD Card Socket
Hinged Lid
FPGA MODES
RESET
R16 10K
R99 0R
R64 10K
R100 0R
R1001 10K
R98 10K
PROGMODE
MODE
EN1
RESET_IN
J5
CONN PCB 3
L29 Ferrite Bead
C66 0.1uF
C67 0.1uF
U16 ADUM1285
ISO_SIGNAL0
ISO_SIGNAL1
16-channel 16-bit DAC
(set VDDA + VDDD TILEx)
C36 0.1uF
C1 10uF
C38 0.1uF
U2 AD5677R
SET_VDDA_TILE1
SET_VDDD_TILE1
SET_VDDA_TILE2
SET_VDDD_TILE2
SET_VDDA_TILE3
SET_VDDD_TILE3
SET_VDDA_TILE4
SET_VDDD_TILE4
SET_VDDA_TILE5
SET_VDDD_TILE5
SET_VDDA_TILE6
SET_VDDD_TILE6
SET_VDDA_TILE7
SET_VDDD_TILE7
SET_VDDA_TILE8
SET_VDDD_TILE8
I2C Address 0001100
Power Input: 48 VDC / 2 A
REVERSE/OVER VOLTAGE PROTECTION @ 60V
Supply +
Supply -
J9-1
J9-2
J9-3
J9-4
F2 500mA
F1 Fuse 2.5A
F3 500mA
D7 S5QB M3G Diode 5A
D1 SMBJ60A TVS 60V
C2 120uF
T1 1 kOhms @ 10MHz
C253 22nF
C251 1uF
C252 22nF
TP1 TESTPOINT
TP2 TESTPOINT
VIN+
VIN-
REVISION RECORD
LTR
ECO NO:
APPROVED:
DATE:
FRONT PANEL & PERIPHERALS
COMPANY: University of California Davis
TITLE: LarPix PACMAN Version 4
DRAWN: H.G.Berns
DATED: 07/21/2021
CHECKED: <Checked By>
DATED: <Checked Date>
QUALITY CONTROL: <QC By>
DATED: <QC Date>
RELEASED: H.G.Berns
DATED: 07/21/2021
CODE: <Code>
SIZE: C
DRAWING NO: PacMan_V4.sch
REV: 4
SCALE: <Scale>
SHEET: 2 of 13



DC-DC Converter
Input: 18V - 75V / 50W
Output: 3.0V / 13.3A / 40W (Input 18V-36V)
Output: 3.0V / 16.6A / 50W (Input 36V-75V)

CCG304803S Output Voltage Adjustment with TRM:

VOUT	R5 (up)	R6 (down)	
3.0 V	DNI	30.7K	
3.1 V	DNI	57.5K	
3.2 V	DNI	137.7K	
3.3 V	DNI	DNI	
3.4 V	81.7K	DNI	
3.5 V	33.4K	DNI	
3.6 V	17.2K	DNI	<-- selected

PKU5510S Output Voltage Adjustment with TRIM:

VOUT	R10 (up)	R11 (down)	
3.0 V	DNI	1.22K	<-- selected
3.1 V	DNI	2.92K	
3.2 V	DNI	8.03K	
3.3 V	DNI	DNI	
3.4 V	4.55K	DNI	
3.5 V	1.43K	DNI	
3.6 V	389R	DNI	

REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

POWER

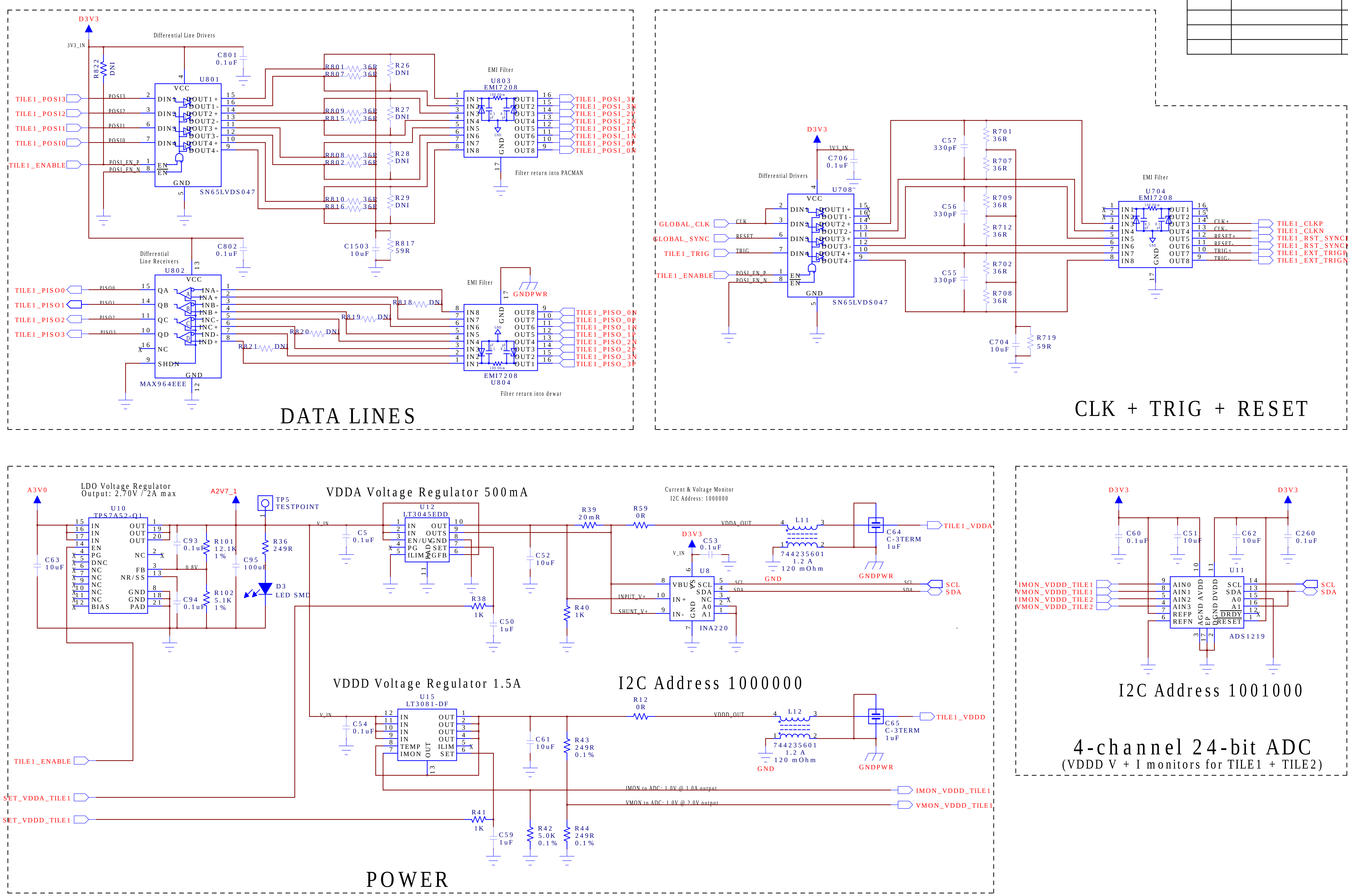
COMPANY: University of California Davis

TITLE: LarPix PACMAN Version 4

DRAWN: H.G.Berns DATED: 07/21/2021
CHECKED: <Checked By> DATED: <Checked Date>
QUALITY CONTROL: <QC By> DATED: <QC Date>
RELEASED: H.G.Berns DATED: 07/21/2021

CODE: <Code> SIZE: C DRAWING NO: PacMan_V4.sch REV: 4

SCALE: <Scale> SHEET: 3 of 13



INA219 + ADS1219 I2C Address Select:

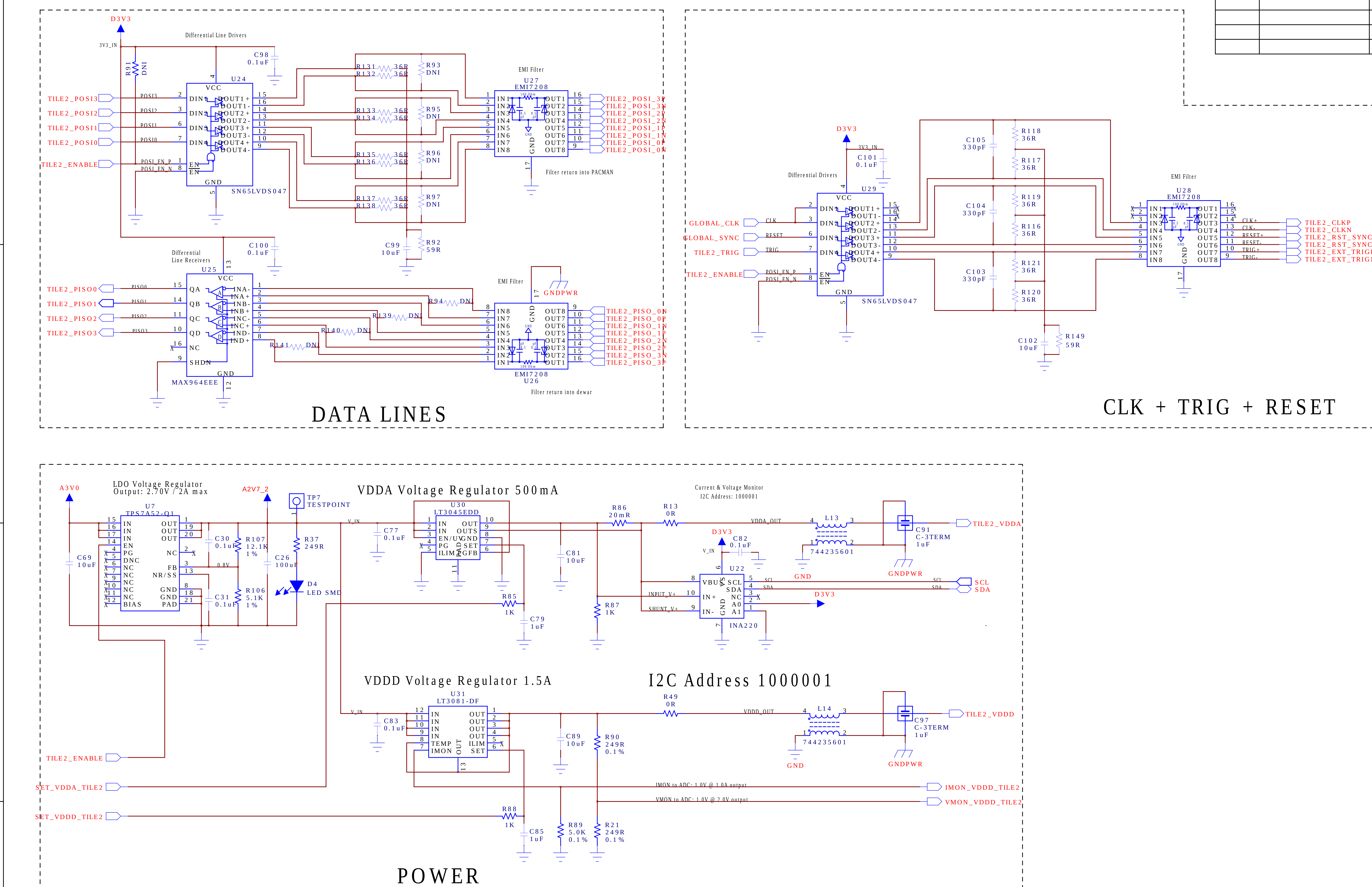
A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
GND	SDA	1000010	VDDA TILE3
GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

TILE1

DRAWN: H.G.Berns	DATED: 07/21/2021	COMPANY: University of California Davis	
CHECKED: <Checked By>	DATED: <Checked Date>	TITLE: LarPix PACMAN Version 4	
QUALITY CONTROL: <QC By>	DATED: <QC Date>	CODE: <Code>	SIZE: C
RELEASED: H.G.Berns	DATED: 07/21/2021	DRAWING NO: PacMan_V4.sch	REV: 4
		SCALE: <Scale>	SHEET: 4 of 13

REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:



REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

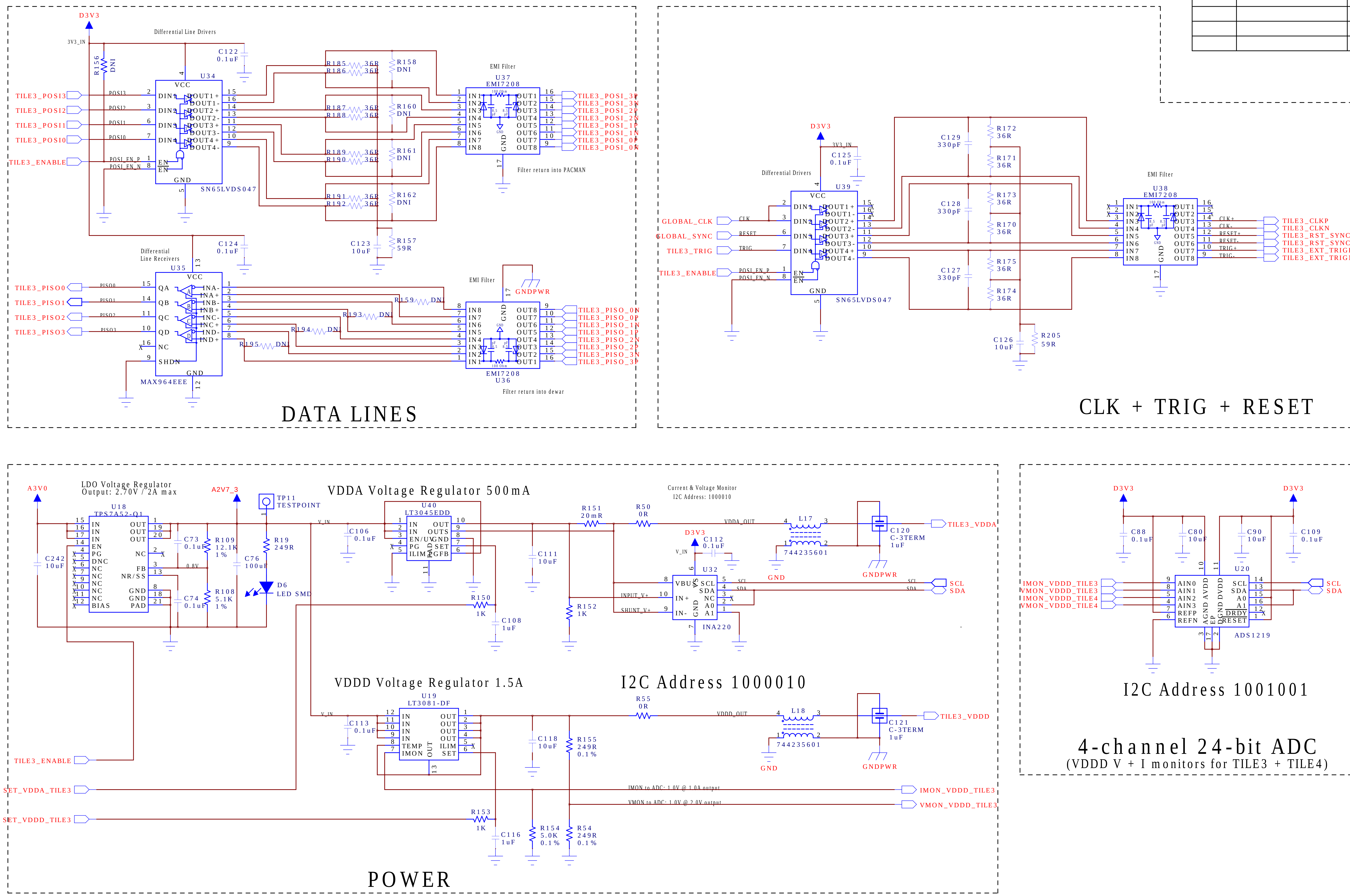
INA219 + ADS1219 I2C Address Select:

A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
GND	SDA	1000010	VDDA TILE3
GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

TILE2

DRAWN: H.G.Berns	DATED: 07/21/2021	COMPANY: University of California Davis			
CHECKED: <Checked By>	DATED: <Checked Date>	TITLE: LarPix PACMAN Version 4			
QUALITY CONTROL: <QC By>	DATED: <QC Date>	CODE: <Code>	SIZE: C	DRAWING NO: PacMan_V4.sch	REV: 4
RELEASED: H.G.Berns	DATED: 07/21/2021	SCALE: <Scale>		SHEET: 5 OF 13	

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



INA219 + ADS1219 I2C Address Select:

A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
GND	SDA	1000010	VDDA TILE3
GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

TILE3

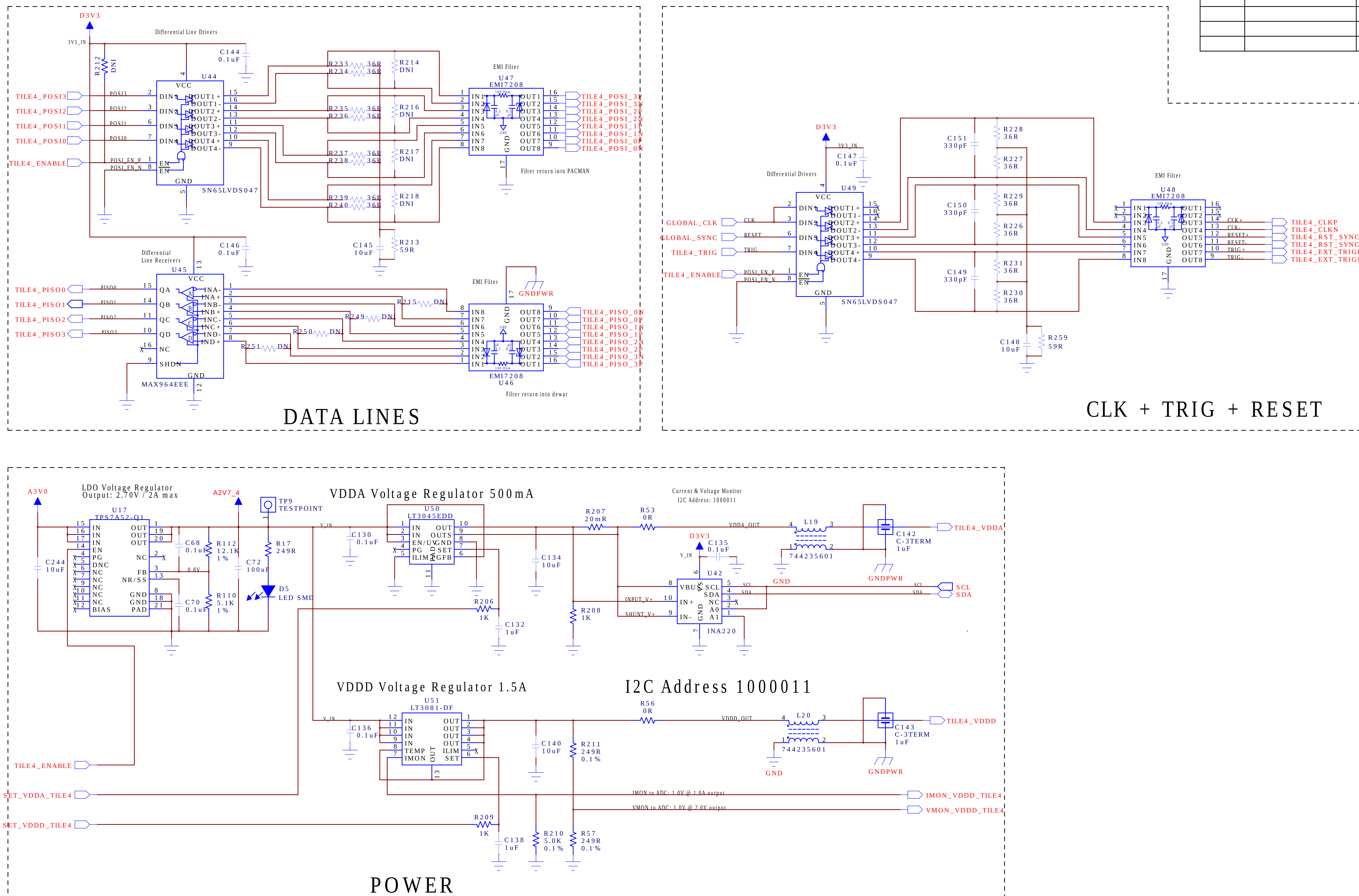
DRAWN: H.G.Berns	DATED: 07/21/2021
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: H.G.Berns	DATED: 07/21/2021

COMPANY: University of California Davis

TITLE: LarPix PACMAN Version 4

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	C	PacMan_V4.sch	4

SCALE: <Scale> SHEET: 6 13



REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

INA219 + ADS1219 I2C Address Select:

A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
GND	SDA	1000010	VDDA TILE3
GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

TILE4

COMPANY: University of California Davis

TITLE: LarPix PACMAN Version 4

DRAWN: H.G.Berns DATED: 07/21/2021

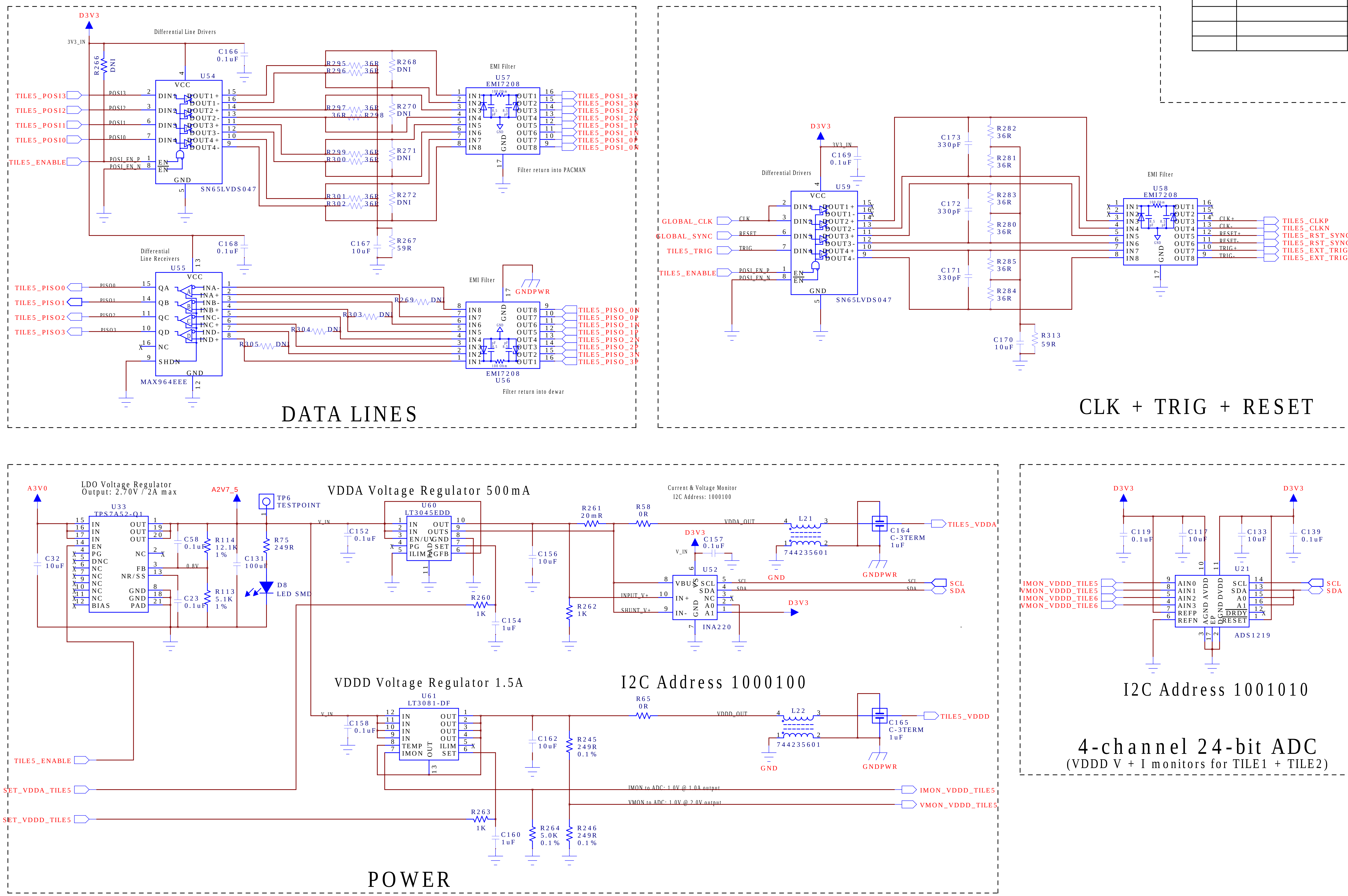
CHECKED: <Checked By> DATED: <Checked Date>

QUALITY CONTROL: <QC By> DATED: <QC Date>

RELEASED: H.G.Berns DATED: 07/21/2021

CODE: <Code> SIZE: C DRAWING NO: PacMan_V4.sch REV: 4

SCALE: <Scale> SHEET: 7 of 13



REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

INA219 + ADS1219 I2C Address Select:

A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
GND	SDA	1000010	VDDA TILE3
GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

TILE5

COMPANY: University of California Davis

TITLE: LarPix PACMAN Version 4

DRAWN: H.G.Berns — DATED: 07/21/2021

CHECKED: <Checked By> — DATED: <Checked Date>

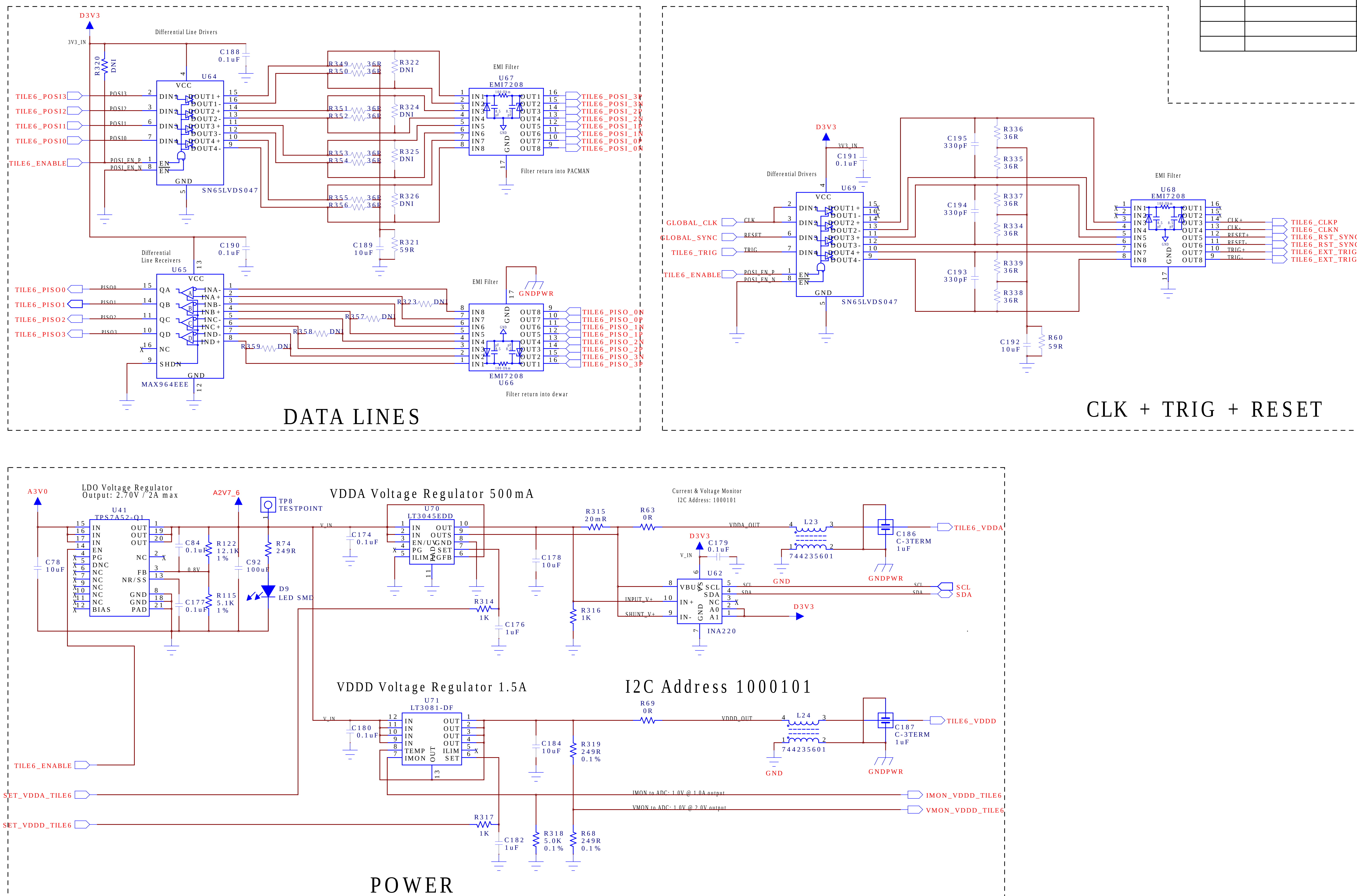
QUALITY CONTROL: <QC By> — DATED: <QC Date>

RELEASED: H.G.Berns — DATED: 07/21/2021

CODE: <Code> | SIZE: C | DRAWING NO: PacMan_V4.sch | REV: 4

SCALE: <Scale> | SHEET: 8 of 13

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

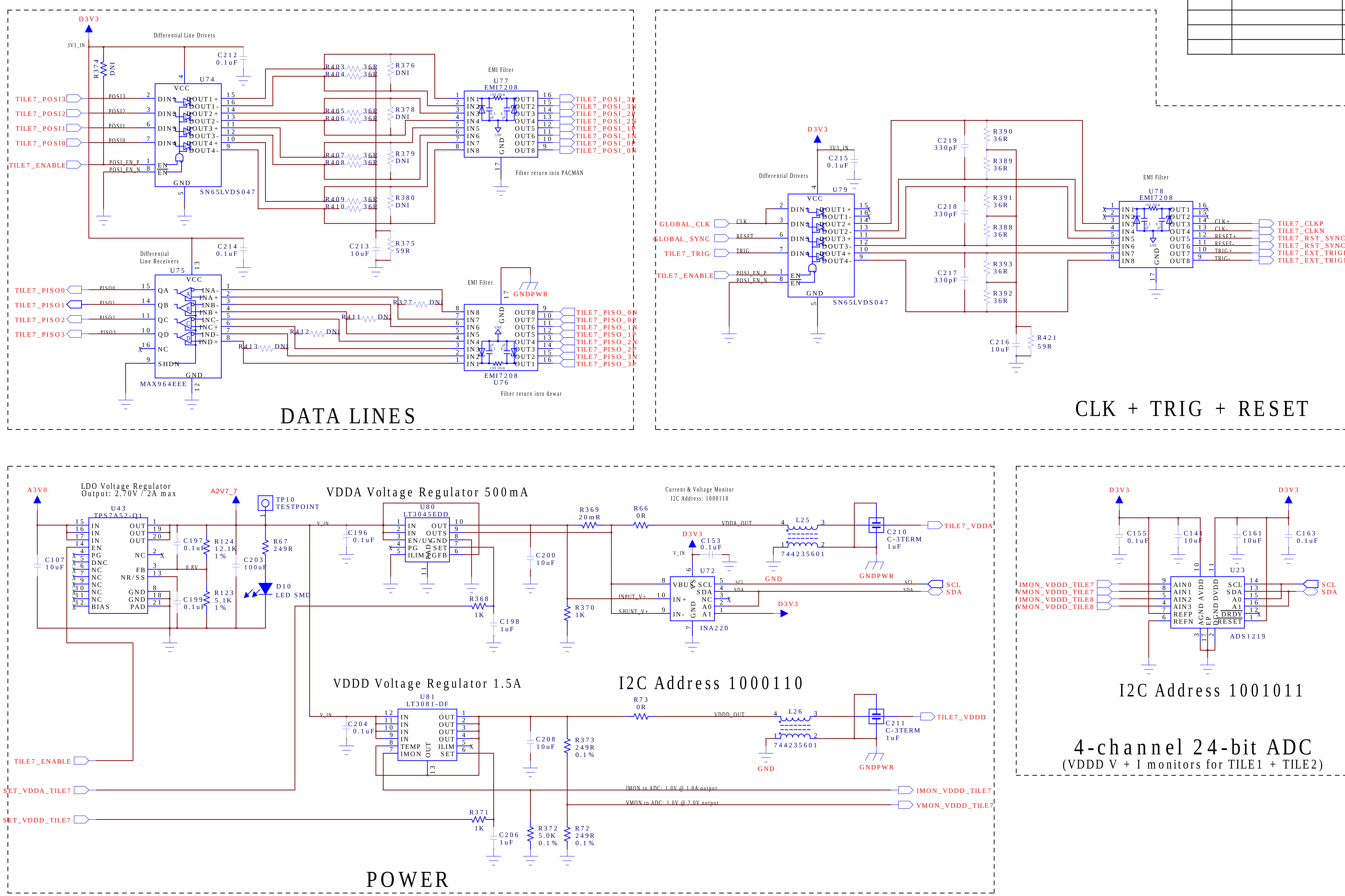


INA219 + ADS1219 I2C Address Select:

A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
GND	SDA	1000010	VDDA TILE3
GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

TILE6

COMPANY: University of California Davis					
TITLE: LarPix PACMAN Version 4					
DRAWN: H.G.Berns	DATED: 07/21/2021	CODE: <Code>	SIZE: C	DRAWING NO: PacMan_V4.sch	REV: 4
CHECKED: <Checked By>	DATED: <Checked Date>				
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
RELEASED: H.G.Berns	DATED: 07/21/2021	SCALE: <Scale>		SHEET: 9 OF 13	



INA219 + ADS1219 I2C Address Select:

A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
GND	SDA	1000010	VDDA TILE3
GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

TILE7

DRAWN: H.G.Berns	DATED: 07/21/2021
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: H.G.Berns	DATED: 07/21/2021

COMPANY: University of California Davis

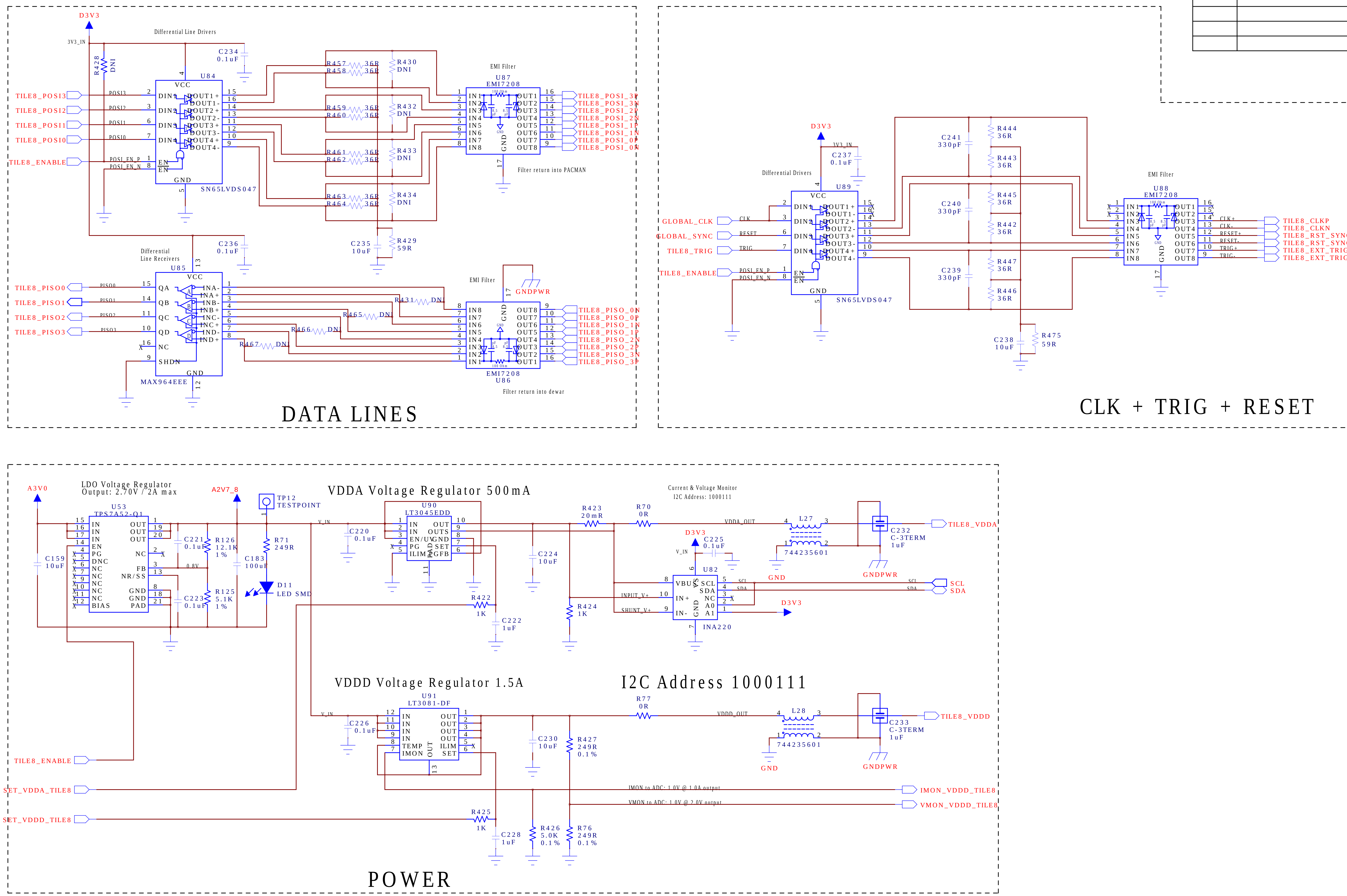
TITLE: LarPix PACMAN Version 4

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	C	PacMan_V4.sch	4

SCALE: <Scale>

LTR	ECO NO:	APPROVED:	DATE:

REVISION RECORD



INA219 + ADS1219 I2C Address Select:

A1	A0	I2C Address	
GND	GND	1000000	VDDA TILE1
GND	VS	1000001	VDDA TILE2
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GND	SCL	1000011	VDDA TILE4
VS	GND	1000100	VDDA TILE5
VS	VS	1000101	VDDA TILE6
VS	SDA	1000110	VDDA TILE7
VS	SCL	1000111	VDDA TILE8
SDA	GND	1001000	VDDD TILE1+TILE2
SDA	VS	1001001	VDDD TILE3+TILE4
SDA	SDA	1001010	VDDD TILE5+TILE6
SDA	SCL	1001011	VDDA TILE7+TILE8
SCL	GND	1001100	(reserved by MAX14661 = 16:2 MUX)
SCL	VS	1001101	
SCL	SDA	1001110	
SCL	SCL	1001111	

TILE8

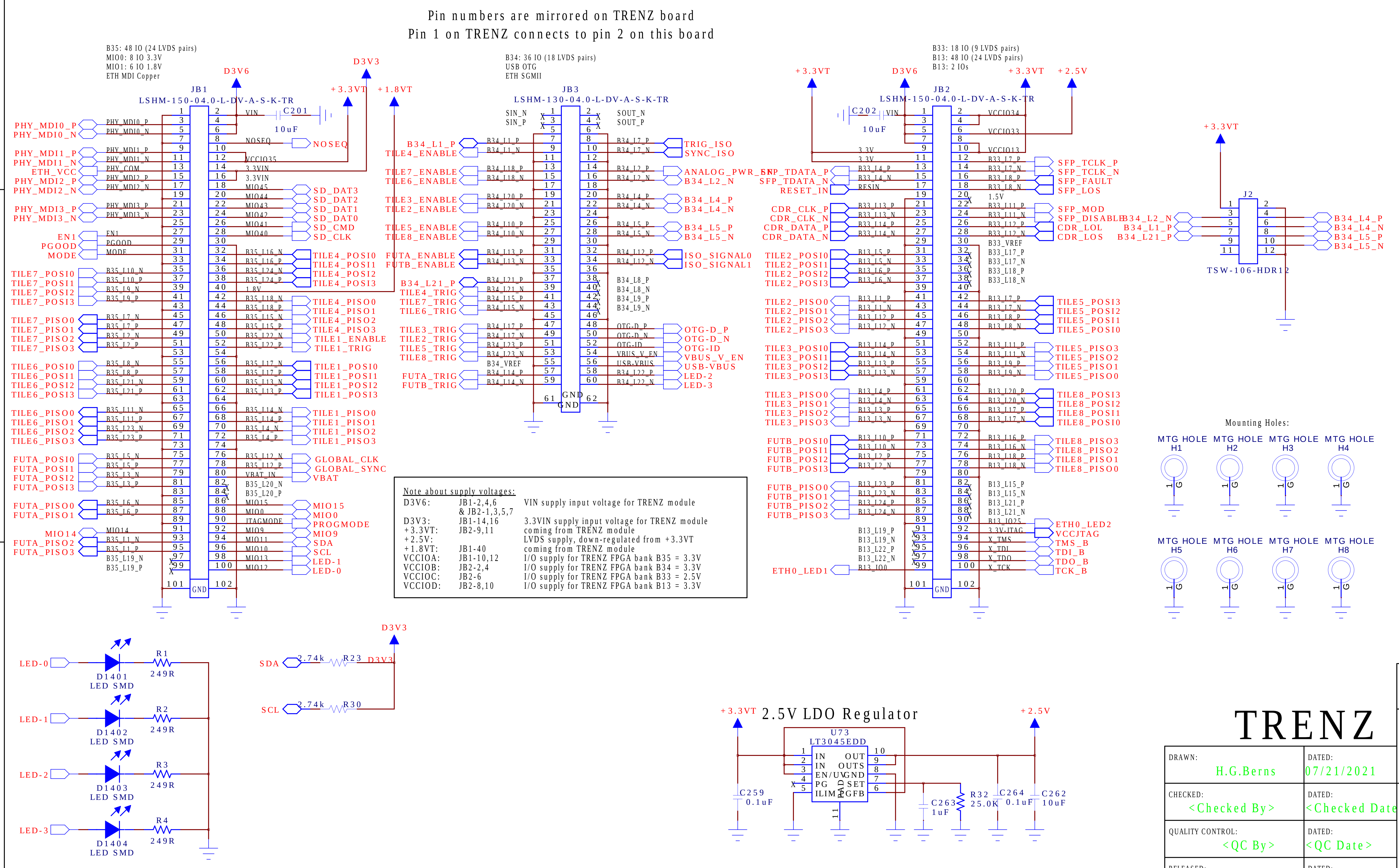
COMPANY: University of California Davis

TITLE: LarPix PACMAN Version 4

DRAWN: H.G.Berns	DATED: 07/21/2021
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: H.G.Berns	DATED: 07/21/2021

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	C	PacMan_V4.sch	4

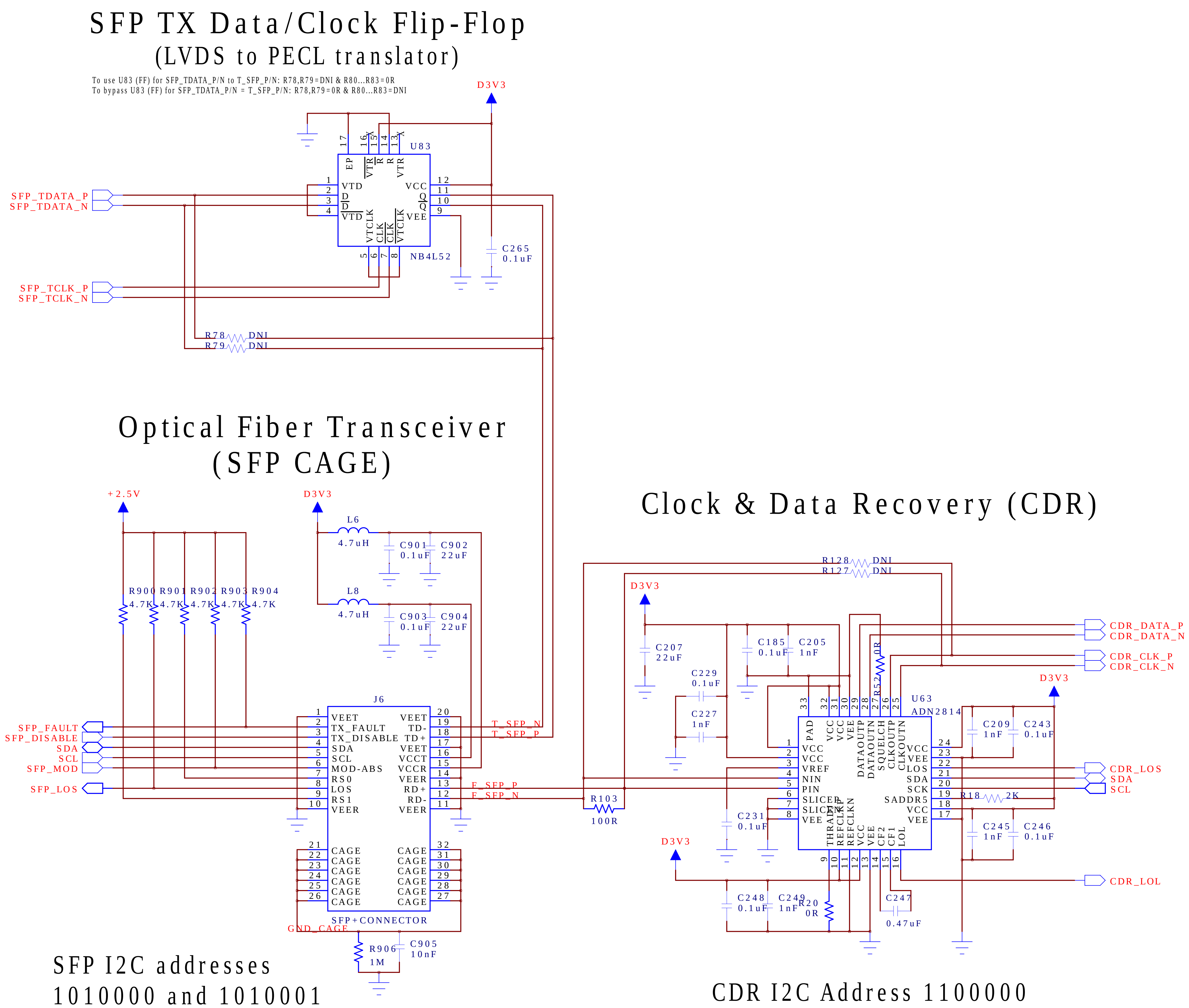
SCALE: <Scale> SHEET: 11 OF 13



REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE:

COMPANY:	University of California Davis
TITLE:	LarPix PACMAN Version 4
DRAWN: H.G.Berns	DATED: 07/21/2021
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: H.G.Berns	DATED: 07/21/2021
CODE: <Code>	SIZE: C
DRAWING NO: PacMan_V4.sch	REV: 4
SCALE: <Scale>	SHEET: 12 of 13



TIMING

COMPANY:	University of California Davis
TITLE:	LarPix PACMAN Version 4
DRAWN: H.G.Berns	DATED: 07/21/2021
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: H.G.Berns	DATED: 07/21/2021
CODE: <Code>	SIZE: C
DRAWING NO: PacMan_V4.sch	REV: 4
SCALE: <Scale>	SHEET: 13 OF 13

REVISION RECORD

LTR	ECO NO:	APPROVED:	DATE: